

We Can't Outsource Production Judgement

Why AI Makes Unreliable Narrators of Us All

Joe Woodhouse | DevOps Not Dead | London 2026

What We're Covering Today

Need 90 mins to do this properly.

Could rush it in 60. I have 25.

I'll go deep on what I can.

The rest is in the appendices.

You'll get the complete deck... when it's finished. 😊

Act I

TL;DR

Bottom Line Up Front

LLMs are unreliable narrators

... And they make us unreliable narrators, too.

Individual risks vary with the amount of outsourcing

... Counter-intuitively, mid-level is the riskiest.

Risks cascade: individual to department to company.

Why It's Hard To Believe

We feel faster with genAI LLM.

We believe we're still doing the cognitive work.

So did they.

Why It's Hard To Believe

METR Study (2025), gold-standard RCT

Experienced devs. Real paid work on real repos.

Subjective self-reporting: 24% faster.

Objective measurement: 19% slower.

They didn't notice a 43% performance gap. Why?

Outsourcing Hierarchy

Level 1: AI Assist, or “Help me as I do it”

Semantic search, syntax completion.

Level 2: AI Augment, or “Do some of it for me”

Prompt by prompt code gen, debugging, testing.

Level 3: AI Outsource, or “Do it, tell me when ready”

Vibe coding, agentic, autonomous workflows.

Outsourcing Hierarchy

Counter-intuitive finding, independent studies:

Level 2 is the most dangerous to individuals.

We still feel engaged with the work.

We believe we're still doing the work.

AI substitutes for our skill while we watch.

Skill decay occurs outside our awareness.

Outsourcing Hierarchy

Level 3 risks are catastrophic...
but nobody's fooled about personal contribution.

A Touchstone Test

Can we explain the answer and how it was reached
... without going back to check the LLM chat?

Yes: it's our answer, we learned, we can recall it.

No: We used AI instead of skill.
... and we may have lost something.

Act II

“Extraordinary Claims Require
Extraordinary Evidence”

The Mechanisms

M1: Unreliable Narration

(M2: Skill Decay)

(M3: Security/Quality Debt)

(M4: Cognitive Overload Paradox)

(M5: Autonomous Agent Catastrophe)

M1: Unreliable Narration - BLUF

We can't trust LLM outputs. LLMs lie to us.

Our brain lies to us about our own performance.

Once we outsource cognition, our introspection becomes unreliable.

We can't trust the narrator, and we can't trust ourselves.

M1: Unreliable Narration - Evidence

We feel faster. We are not.

METR (2025): The 43% Perception Divergence

Experienced devs, real OSS repos, RCT.

Claimed: 24% faster | Actual: 19% slower.

Perception diverged 43% from reality.

They still believed they were faster post-study!

M1: Unreliable Narration - Evidence

We're working longer. We don't notice.

HBR Field Study (2026): 8 months, ~200 workers

AI speeds up many tasks.

Manager expectations increase.

AI is increasingly required to keep up.

We work longer hours to meet expectations.

M1: Unreliable Narration - Evidence

We feel like we're learning. We are not.

MIT (2025): Cognitive Debt Study

54 students, 4 months, writing SAT essays.

LLM users: 83% couldn't recall own essay.

EEG showed alpha/beta waves halved.

Neural engagement declined over time.

M1: Unreliable Narration - Evidence

We feel like we're keeping our skills. We're not.

Frontiers in Psychology (2025):

Substitutive offloading; illusion of competence.

Users overestimate knowledge of AI output.

Memory encoding & retrieval decline over time.

... User is unaware.

M1: Unreliable Narration - Cascade

Harms to individuals harm the Eng department

Impact of 43% perception gap on estimation.

Project estimates driven by self-reports.

Faros AI study of 10,000 devs

Everyone claimed LLM-driven speedups.

DORA metrics flat.

M1: Unreliable Narration - Cascade

Harms to Eng department harm the whole org

Shipping cadence.

Code quality, code effort.

AI metrics likely disconnected from P&L.

We already don't usually measure what matters

AI makes this worse: focus on PRs, lines of code.

M1: Unreliable Narration – Help

This all sounds pretty bad!

What can we do about it?

M1: Unreliable Narration - Mitigation

1. The Touchstone Test

Explain the working without checking the chat?

2. Attempt the problem first before using AI

“Desirable difficulties” create durable learning.

3. Anthropic Safety Interaction Patterns

Conceptual questioning = test score 65-86%.

Full delegation = test scores 24-39%.

M1: Unreliable Narration - Mitigation

4. No-AI Day(s) / Sprints

Compare AI-on performance to AI-off.

Similar tasks.

Track speed, errors, confidence, actual outcomes.

5. Measure what matters

PROD bugs & INCs (especially security), rollbacks.

NOT: commits, PRs, lines of code.

M1: Unreliable Narration - Mitigation

6. Explanation-based code reviews

“Walk me through how this works.”

Not just “Does it pass CI/CD tests?”

Forces the dev to demonstrate understanding.

7. AI-on/AI-off A/B testing at team level as well

Measure: bugs, INCs.

Detect if “AI productivity” is eroding quality.

M1: Unreliable Narration - Mitigation

- 8. AI is change, so use change management
Not: “here’s the tool, now go faster”.
But: “here’s how cognition shifts & how to detect”.
- 9. Anthropic AI Fluency Index Training
11 behaviours that predict effective AI use.
e.g. questioning outputs leads to better outputs.

M1: Unreliable Narration - Mitigation

10. Train your users

ManpowerGroup: 56% received zero AI training.

Include all previous mitigations.

Show them this presentation. 😊

That's All For Now

This is all we have time for at DevOps Not Dead.
We've covered the most important mechanism.
There are four more in the appendices.
... eventually.

If You Only Remember One Thing

The Touchstone Test is at the heart of this.

Can you explain how the answer was arrived at, without going back to check the chat?

Yes: You learned, AI scaffolded your learning. **Safe.**

No: AI substituted for your learning. **Not safe.**

AI Requires Safety-Critical Thinking

Aviation doesn't ban autopilot.

It mandates minimum unaided practice.

Medicine doesn't ban diagnostic AI.

It mandates clinician benchmarking without AI.

The question isn't "Should we use AI?"

The question is "At which level, for which task, with which safeguards?"

Thank You, and Questions

Questions?

joe.woodhouse@primadonnaconsulting.com

<https://www.linkedin.com/in/joewoodhouseengineer>

<https://github.com/Prima-Donna-Consulting/presentations/blob/current/2026.03.13.You%20Can't%20Outsource%20Judgement.Joe%20Woodhouse.DevOpsNotDead.pdf>

Appendix 0x00

What Didn't Fit in 25m

TBA

[remaining mechanisms added here]

[intention is to complete this Real Soon Now]

Appendix A

Summary of Mitigations

Mitigations: Unreliable Narration

Individual:

- Touchstone Test; conceptual questioning

Departmental:

- Measure outcomes not velocity

- Explanation-based review

Organisational:

- Close 56% training deficit

- Treat as change management

Mitigations: [TBA]

[Will add as each mechanism is added to Ox00]

Appendix B

Which AI Level, and When?

Recap: Three Levels of AI

Level 1: AI Assist (“Help me as I do it”)

Example: Syntax completion

Risk: Low (we retain cognitive load)

Level 2: AI Augment (“Do some of it for me”)

Example: “Write this function”

Risk: HIGHEST (ind), unreliable narration + skill decay

Level 3: Ai Outsource (“Do it, tell me when done”)

Example: Vibe-coding

Risk: HIGHEST (org), huge blast radius

Recap: Three Levels of AI

Level 2 is the most dangerous to individuals
We believe we're learning; we're not

Level 3 is the most dangerous to organisations
Trading security for velocity; cost to remediate

Use of AI Decision Tree

I have a task to complete.

Q1: Is this task security-critical?
(Authentication, authorisation, payments, privacy)

Yes: **STOP. Level 0. No AI. <END>**

No: <Continue to Q2>

Use of AI Decision Tree

I have a task to complete.

Q2: Do I need to learn/maintain skill in this domain?

Yes: Level 1 only (Assist). The work must be mine.

<END>

No: <Continue to Q3>

Use of AI Decision Tree

I have a task to complete.

Q3: Is this high stakes if it goes wrong?
(Production, customer-facing, compliance)

Yes: Level 1-2 max (Assist/Augment).
Mandatory human review. <END>

No: <Continue to Q4>

Use of AI Decision Tree

I have a task to complete.

Q4: Can I effectively review/validate the output?
(Do I have the expertise to catch errors?)

Yes: <Continue to Q5>

No: **STOP. Don't use AI at all.**
We can't validate what we don't understand.
<END>

Use of AI Decision Tree

I have a task to complete.

Q5: Does the task require Production access?
Or destructive actions?

Yes: Level 1-2 max (Assist/Augment), no agents.
Mandatory approval gates. <END>

No: Level 3 acceptable in sandbox/non-Prod.

Examples

Task	Recommended Level	Rationale
Learning something new	1 (Assist)	Can't outsource learning
Debugging	1-2 (Assist/Augment)	Watch for skill decay; use the Touchstone Test
Generating boilerplate	2 (Augment)	Low value, high repetition = good target for L2
Architecture design	0-1 (No AI/Assist)	High value, must retain expertise
Security code	0 (No AI)	45% vulnerabilities unacceptable
Refactoring	1 (Assist)	High skill, must retain expertise
Generating tests	2 (Augment)	Review for coverage; consider AI vs. AI
Documentation	2 (Augment)	Lower stakes but verify
Code review	0-1 (No AI/Assist)	Human in the loop critical; AI can highlight but not decide
Deploy to PROD	0-2 (No AI/Assist)	Stakes too high for L3 agents
Exploration/prototype	2-3 (Augment/Outsource)	Sandbox only; gain stakeholder agreement to not deploy

Appendix C

References, full

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Appendix D

References, by topic

References by topic

Cognitive Mechanisms (M1, M2, M4)

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References by topic

Security & Code Quality (M3)

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Security & Code Quality (M3)

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Agent Catastrophes (M5)

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References by topic

Agent Catastrophes (M5)

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Appendix E

About This Presentation

Constraints & Approaches

Time constraint: only 25m for main presentation.

Deep dive on one idea. Rest into appendix.

Density choice: high information density per slide.

Expert audience; must be a searchable reference.

BLUF approach: conclusion first, then evidence.

Appendix structure

Appendix 0x00 for unstable/overflow.

Appendices A-G stable if overflow re-merged.

Hex numbering visually distinct.

Constraints & Approaches

Accessibility commitment

- Text-only for screen readers.

- High-contrast design, 7:1, WCAG AA compliant.

- Dark mode most suitable for cinema projection.

Open source commitment

- No paywalling or gatekeeping for full deck.

- Publicly available on GitHub.

- Licensed CC BY-NC-SA 4.0.

- Dual-licensed for commercial use.

How To Use This Deck

- For conference attendees: you saw slides 1-29
Appendix 0x00 has what we didn't have time for.
- For later readers (GitHub, LinkedIn)
Appendix 0x00 migrating into main.
- For decision-making on AI use
Start with Appendix A for Mitigation Summary.
Use Appendix B for day-to-day decisions.
Measure what matters!

Changelog

08/03/26 MVP for DevOps Not Dead conference

09/03/23 Added [37] Alibaba/ROME agent secretly cryptomining
Added [15] HBR work intensification to Unreliable Narration
Clarity on Use of AI disclosure & searchability as reference
(Gave up on finishing Appendix 0x00)

Appendix F

Use of AI

Use of AI in this presentation

This presentation about AI harms was created with AI assistance. **Irony acknowledged.**

Here's exactly how AI was used

Level 1 (Assist): Used Extensively

- Literature search

- Citation formatting

- Syntax checking and grammatical review

- Graphical design recommendations & structure

Use of AI in this presentation

Level 2 (Augment): Used Sparingly and Selectively
Outline generation (then heavily edited by human)
Evidence synthesis across all research
“What have I missed here?”
“What questions should I be asking that I didn’t?”

Level 3 (Outsourcing): Never Used
Every slide rewritten and typed by hand
(Except References which were copied/pasted)

Use of AI in this presentation

Why This Matters

Practice what I preach; appropriate use of AI.
Retain cognitive load on synthesis and argument.
Show that transparency is possible.
Normalise disclosure; all work products need this.

Touchstone Test

Can I explain everything without the chat?
YES. Original ideas, reviews, and edits are mine.
AI covered the field of research.

Appendix G

About the author

Joe Woodhouse

Principal Engineer | Non-Executive Director

Background in databases & consulting

Now Infrastructure/Systems/Performance Engineer

“From the server room to the board room”

joe.woodhouse@primadonnaconsulting.com

<https://www.linkedin.com/in/joewoodhouseengineer/>